Summary

A clear understanding of the foundations of the physical world forms a solid basis for further study of physics. Understanding of the relationship between the states of matter helps to explain different types of everyday physical changes that we see around us.

Underlying knowledge and understanding

Learners should be familiar with the structure of matter and the similarities and differences between solids, liquids and gases. They should have an idea of the particle model and be able to use it to model changes in particle behaviour during changes of state. Learners should be aware of the effect of temperature in the motion and spacing of particles and an understanding that energy can be stored internally by materials.

Common misconceptions

Learners commonly carry misconceptions about matter: assuming atoms are always synonymous with particles. Learners also struggle to explain what is between the particles, instinctively ‘filling’ the gaps with ‘air’ or ‘vapour’. They often struggle to visualise the 3D arrangement of particles in all states of matter. Learners can find it challenging to understand how kinetic theory applies to heating materials and how to use the term temperature correctly, regularly confusing the terms temperature and heat.

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
PM1.2i	apply: change in thermal energy (J) = mass (kg) × specific heat capacity (J/kg °C) × change in temperature (°C)	M1a, M3b, M3c, M3d
PM1.2ii	apply: thermal energy for a change in state (J) = mass (kg) × specific latent heat (J/kg)	M1a, M3b, M3c, M3d

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
P1.2a describe how mass is conserved when substances melt, freeze, evaporate, condense or sublimate			WS1.3a, WS1.3e, WS1.4a, WS2a, WS2c	Use of a data logger to record change in state and mass at different temperatures. (PAG P5) Demonstration of distillation to show that mass is conserved during evaporation and condensation. (PAG P5)

P1.2b	describe that physical changes differ from chemical changes because the material recovers its original properties if the change is reversed				
P1.2c	describe how heating a system will change the energy stored within the system and raise its temperature or produce changes of state			WS1.3a, WS1.3e, WS1.4a, WS2a, WS2b, WS2c	Observation of the crystallisation of salol in water under a microscope. Use of thermometer with a range of -10–110 °C, to record the temperature changes of ice as it is heated. (PAG P1)
P1.2d	define the term specific heat capacity and distinguish between it and the term specific latent heat	specific latent heat of fusion and specific latent heat of vaporisation		WS1.2e, WS1.3b, WS1.3c, WS1.3h, WS1.4a, WS1.4f, WS2a, WS2b	Investigation of the specific heat capacity of different metals or water using electrical heaters and a joulemeter. (PAG P5)
P1.2e	apply the relationship between change in internal energy of a material and its mass, specific heat capacity and temperature change to calculate the energy change involved		M1a, M3c, M3d		
P1.2f	apply the relationship between specific latent heat and mass to calculate the energy change involved in a change of state		M1a, M3c, M3d	WS1.2e, WS1.3b, WS1.3c, WS1.3h, WS1.4a, WS1.4f, WS2a, WS2b	Measurement of the specific latent heat of vaporisation of water. (PAG P5) Measurement of the specific latent heat of stearic acid. (PAG P5)